

SAT UZBEKISTAN

Test creator Xodiyev Lutfulloxon

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1

If $\frac{2x-3}{2} = k-1$ and $k=5$, what is the value of $2x$?

- A) 4
- B) 5.5
- C) 8
- D) 11

2

$$(5+3i) - (8-2i) = a+bi$$

In the equation above, a and b are real numbers.

If $i = \sqrt{-1}$, what is the value of b ?

- A) -1
- B) 1
- C) -5
- D) 5

3

If Claire paid k dollars for a computer that was only 20 dollars more than half the original price, what was the original price, in dollars?

- A) $k+20$
- B) $k-40$
- C) $2k-20$
- D) $2k-40$

4

Jenny is on the school swim team and has swim practice m hours in the morning and p hours in the evening each day. The schedule is the same each day. If she swims a total of k hours for five days, which of the following is the expression for m ?

- A) $\frac{k-p}{5}$
- B) $\frac{k-5p}{5}$
- C) $k-5p$
- D) $5(k-p)$

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A certain business is marketing its product and has determined that, when it raised the selling price of its product, its sales went down. The number of units sold, P , is modeled by the equation $P = 1200 - 20s$, where s is the selling price, in dollars. Based on this model, what is the decrease in selling price from 700 units sold to 900 units sold?

- A) 5
- B) 10
- C) 15
- D) 20

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$$(x^2 + y^2)^2 - (x^2 - y^2)^2$$

Which of the following is equivalent to the expression above?

- A) $x^4 - y^4$
- B) $2(x^2 + y^2)$
- C) $2x^2y^2$
- D) $4x^2y^2$

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Kimberly earns k dollars per week. At this rate how many weeks will it take her to earn p dollars?

- A) $\frac{p}{k}$
- B) $\frac{k}{p}$
- C) kp
- D) $\frac{10p}{k}$

8

If $\frac{2a}{b} = 5$, what is the value of $\frac{5b}{a}$?

- A) 2
- B) 4
- C) 10
- D) 12.5

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$$2x + by = 10$$

$$ax + 4y = 15$$

In the system of equations above, a and b are constants and $a = 2b$. If the system has no solution, which of the following could be a possible value of a ?

- A) -2
- B) $\frac{1}{2}$
- C) 4
- D) 8

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$$f(x) = ax^2 - 15$$

For the function f defined above, a is a constant and $f(3) = 10$. Which of the following is equal to the value of $f(5)$?

- A) $f(0)$
- B) $f(3)$
- C) $f(-3)$
- D) $f(-5)$

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A certain job can be done in 20 hours by 4 people. How many people are needed to do the same job in 10 hours?

- A) 2
- B) 4
- C) 8
- D) 10

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Which of the following is equivalent to $f(x) = x^2 - 6x + 7$?

- A) $f(x) = (x + 3)^2 + 5$
- B) $f(x) = (x - 3)^2 + 2$
- C) $f(x) = (x - 3)^2 - 2$
- D) $f(x) = (x - 7)(x + 1)$

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If $24x^2 - kx + 16 = (3x + 4)(ax - b)$ for all values of x , where a, b , and k are constants, what is the value of k ?

- A) -44
- B) -12
- C) 12
- D) 44

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In the xy -plane, the equation of line ℓ is $x + 3y = 5$. If line m is perpendicular to line ℓ , what is a possible equation of line m ?

A) $y = -\frac{1}{3}x + 2$

B) $y = \frac{1}{3}x - 1$

C) $y = -3x + 1$

D) $y = 3x + \frac{2}{3}$

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If $a + b = 8$ and $\frac{27^a}{3^b} = 81$, what is the value of a ?

A) 3

B) 4

C) 5

D) 6

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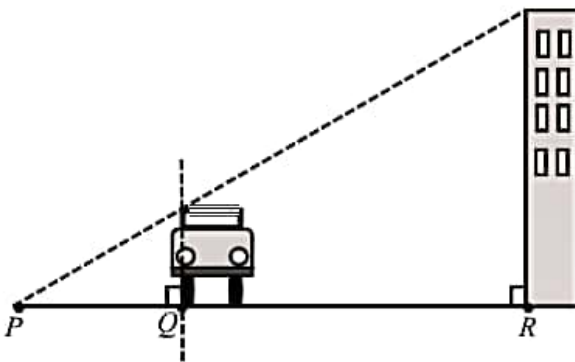
16

In a right triangle, one of the angles is x° . If $\tan x^\circ = \frac{5}{12}$, what is the value of $\sin x^\circ$?

- A) $\frac{3}{16}$ B) $\frac{5}{16}$ C) $\frac{2}{5}$ D) $\frac{3}{32}$

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Dawson needs to measure the height of a building near his house. He chooses a point P on the ground where he can visually align the roof of his car with the edge of the building roof. The height of the car is 4 feet and the distance from point P to point Q is 10 feet, as shown in the figure above. If the distance from point Q to point R is 80 feet, and the height of the building is k feet, what is the value of k ?

- A) 24 B) 36 C) 48 D) 60

18

If $a(x+1) + b(x-1) = 7x$ for all real number x , where a and b are constants, what is the value of a ?

- A) 1.5 B) 3.5 C) 4.5 D) 7

19

According to the formula $p = \frac{4}{3}k + 81$, if the value of p is increased by 16, by how much does the value of k increase?

- A) 9 B) 12 C) 15 D) 18

20

$$x^2 + y^2 = 56$$

$$y = \sqrt{x}$$

According to the system of equations above, what is the value of x ?

- A) 5 B) 7 C) 9 D) 4

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